INVESTMENT MEMORANDUM

SA Risk Solutions

Chargeback Prevention Engine · AI-Powered Fraud Risk Scoring

Confidential · June 2026 · For Discussion Purposes Only

$117B	6.6%	0.63	4
Global chargeback losses (2023)	Chargeback rate in training data	Model ROC-AUC score	Intervention tiers

Executive Summary

SA Risk Solutions has developed an end-to-end, production-ready chargeback prediction engine that scores card transactions in real time for chargeback risk. Built on a Gradient Boosting classifier trained across 10,000 labelled transactions, the engine ingests six categories of transaction signals, outputs a 0–100 risk score, and routes each transaction to one of four intervention tiers — from no action to a merchant hold. A live Streamlit demonstration is publicly accessible, and the full codebase is version-controlled on GitHub.

Market Opportunity

Chargebacks represent one of the most costly and growing problems in global payments. Merchants lose an estimated **$3.75 for every $1 of disputed transaction value** once fees, operational costs, and lost goods are factored in. Friendly fraud — where legitimate cardholders deliberately dispute genuine purchases — now accounts for over 70% of all chargebacks, making behavioural prediction (rather than pure fraud detection) the critical differentiator.

Segment	TAM Estimate	Key Driver
Global Chargeback Losses	$117B annually	E-commerce growth + CNP fraud
Fraud Prevention SaaS Market	$38B by 2027	Real-time decisioning demand
Friendly Fraud Specifically	$50B+ annually	Post-pandemic behaviour shift

Strengths and Weaknesses

■ STRENGTHS	■■ WEAKNESSES
• End-to-end pipeline — data ingestion, feature engineering, model training, scoring, and UI are all production-ready. • Explainable outputs — every score includes a ranked list of risk drivers, meeting regulatory requirements for AI transparency. • Modular architecture — each component (features, model, scorer) is independently testable and replaceable. • Live public demo — a deployed Streamlit app allows immediate stakeholder demonstration without setup. • Four-tier intervention system aligned with real-world payment operations workflows. • Class imbalance handled correctly — a common failure point in fraud models, addressed via sample weighting. • 21 unit tests covering core logic — demonstrates engineering rigour beyond typical portfolio projects.	• ROC-AUC of 0.63 reflects the inherent difficulty of friendly fraud prediction; production systems typically combine ML with additional signals (network graphs, device fingerprinting). • Training data of 10,000 rows is small by industry standards — production deployment would require millions of transactions. • No real-time data pipeline — the current system scores on demand rather than ingesting a live transaction stream. • Single model architecture — production systems typically use ensemble stacking or champion/challenger model frameworks. • No feedback loop — chargebacks confirmed weeks later are not yet fed back to retrain the model automatically.

Commercial Viability

The chargeback prevention market is well-established, with proven willingness to pay from both acquiring banks and large merchants. The SA Risk Solutions engine addresses a well-defined, quantifiable pain point where ROI is directly measurable — every chargeback prevented saves the merchant 3–4x the transaction value in fees and penalties.

The product's explainability features make it particularly suited to regulated environments where black-box AI decisions are increasingly unacceptable. The four-tier intervention system maps directly to existing payment operations workflows, minimising integration complexity for prospective clients.

Potential Customers

Customer Segment	Priority	Value Proposition	Example Names
Acquiring Banks	HIGH	Banks that process card payments on behalf of merchants carry chargeback liability and pay Visa/Mastercard penalty fees when rates exceed thresholds. A prevention engine reduces this directly measurable cost.	Barclays, HDFC Bank, JPMorgan Chase Merchant Services
Payment Processors	HIGH	Companies like Stripe, Adyen, and Razorpay must protect their merchant customer base from chargebacks to reduce platform risk and improve merchant retention.	Stripe, Adyen, Razorpay, PayU, Cashfree
Large E-Commerce Merchants	HIGH	Marketplaces and direct-to-consumer brands with high transaction volumes face disproportionate chargeback losses, particularly in digital goods and subscription categories.	Amazon Marketplace, Flipkart, Myntra, Nykaa
Subscription Businesses	MEDIUM	SaaS, streaming, and subscription box companies face systematic friendly fraud from customers who dispute recurring charges after use.	Netflix, Spotify, gaming platforms, B2B SaaS

Travel Platforms	MEDIUM	Travel has one of the highest chargeback rates due to cancellations, non-delivery of services, and deliberate abuse of refund policies.	MakeMyTrip, Booking.com, airlines, OTAs
Insurtech & BNPL	MEDIUM	Buy Now Pay Later providers and digital insurers face novel chargeback patterns not covered by traditional fraud tools.	LazyPay, ZestMoney, Slice, Digit Insurance

How Payment Companies Can Monetise This

SaaS API Licensing Payment processors and banks integrate the scoring API into their authorisation pipeline. Pricing is per-call, making it frictionless to adopt and directly tied to usage. At 1 billion transactions per year (mid-size processor scale), this generates $1M–$5M ARR from a single client.	**$0.001 – $0.005 per transaction scored**
Risk-Based Interchange Premium Visa and Mastercard already charge issuers and acquirers differently based on transaction risk. A prevention engine can justify charging merchants a small premium on transactions actively protected — framed as a chargeback guarantee or protection product.	**2–5 bps on protected transaction value**
Chargeback Guarantee Product Take on the chargeback liability for a merchant in exchange for a monthly fee and a percentage of prevented losses. The engine's scoring enables selective underwriting — only guarantee transactions the model flags as low risk, creating a profitable spread between premiums collected and losses paid.	**Monthly SaaS + performance fee**
White-Label Platform Licence the full technology stack to regional banks and fintech platforms who want to offer chargeback prevention as their own branded product without building the ML infrastructure themselves.	**$50K – $500K annual licence**
Consulting + Implementation For institutions that want the model customised on their own proprietary transaction data. This is particularly attractive to large acquirers in emerging markets who lack internal data science teams.	**Project-based, $20K – $200K**

Competitive Landscape

Established players include Kount (Equifax), Signifyd, Riskified, and Chargebacks911. SA Risk Solutions differentiates on **explainability**, **open architecture**, and **cost** — enterprise chargeback tools can cost $100K+ annually; an API-first model enables much lower entry pricing for mid-market merchants and regional banks in emerging markets (India, SEA, LatAm) currently underserved by Western incumbents.

Recommended Next Steps

1. Pilot with a regional acquiring bank or payment processor on live transaction data to validate ROC-AUC uplift over their current rules engine.

2. Build a real-time API wrapper (FastAPI) to enable direct integration into authorisation pipelines.

3. Expand training dataset via data partnership with a processor to reach 1M+ transactions.

4. Develop a feedback loop — ingest confirmed chargeback outcomes to retrain the model monthly.

5. File provisional patent on the four-tier intervention routing methodology.
